INSIDE STUDIO SP · VST3 · AU

MERCÚRIO

4-OPERATOR FM SYNTHESIZER

Step-by-step manual

Version 0.6.0 · Windows and macOS · VST3 and AU



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What is MERCÚRIO

MERCÚRIO is a **4-operator**, **16-voice FM** synth in the spirit of the 80s desktop FM boxes, leaning towards **plastic, metallic and psychedelic** sounds: inharmonic ratios, feedback, ring mod and wavefolding. Everything sits on a single screen, no pages.

- **8 algorithms** (who modulates whom), drawn on screen.
- **4 operators**, each with 8 waveforms and its own ratio, envelope and velocity.
- **2 filters** (including COMB and VOWEL), saturation between them, 2 envelopes, 3 LFOs and a 6-slot modulation matrix.
- **Arpeggiator** and an **effects** chain: distortion, analog boost, chorus, phaser, delay, reverb and EQ.
- **360 presets** in 9 categories, level-matched: switching presets causes no jumps in volume.

It's light on the CPU: modulation runs in blocks, and only the operators, filters and gain run per sample.

Before you start

ITEM	DETAILS
Format	VST3 instrument (Windows and macOS) and AU (macOS).
Voices	Up to 16, with POLY, MONO and LEGATO modes.
MIDI	Pitch bend, mod wheel, aftertouch and sustain (CC 64).
Window	From 60% to 180% (opens at 85%), chosen in the preset menu and saved with the project.

Installation and activation

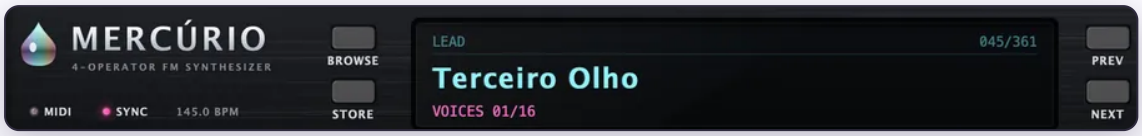
1. Download the installer from the link in your purchase e-mail and choose Windows or Mac.
2. Run the installer and open your DAW. If needed, have it scan for new plugins (rescan).
3. Load MERCÚRIO on an **instrument track** (MIDI).
4. The first time, enter the **e-mail** and the **license token** you received.

> **LICENSE.** The same token works on up to 2 computers. Deactivating it on one computer frees the slot for another. To view the license: `BROWSE` › `Licença…` (License…).

> **UNTIL ACTIVATED**, the instrument stays silent.

STEP 1

1 Pick a preset



Top bar: MIDI and SYNC, BROWSE / STORE, preset display, PREV / NEXT.

- `BROWSE` opens the menu: bank › category › preset. `PREV` and `NEXT` step through the list.
- There are 360 presets in 9 categories: *BASS, LEAD, PAD, PLUCK, BELLS, KEYS, ARP, TEXTURA* (texture) and *FX*, plus Init (a pure sine). The ARP presets come with the arpeggiator already on.
- The display shows bank and category, the position in the list, the name and, for 2.5 s, the last parameter you touched.
- The `MIDI` and `SYNC` LEDs light up when MIDI comes in and when the DAW is playing.

> **TIP.** Each category has 8 different sound families, with 5 variations each. Browse the whole category before you start editing: it's faster to start from a sound close to what you want.

STEP 2

2 FM in 1 minute

In FM, an operator (an oscillator with its own envelope) can either **sound** or **modulate** another one:

- **Carrier (OUT)**: it's what you hear.
- **Modulator (MOD)**: it isn't heard on its own; it changes the timbre of the operator it modulates. The higher the modulator's **LEVEL**, the brighter and richer the sound.
- The modulator's **RATIO** sets the character: whole-number ratios (×1, ×2, ×3) sound harmonic; fractional ratios (×1.41, ×3.5) sound **metallic**, like bells.

The **algorithm** says who modulates whom. In the ALGORITHM display, a **filled** box is a carrier and an **outlined** box is a modulator; the loop on OP 4 is the feedback.

STEP 3

3 FM: algorithm and character



ALGORITHM	WHO MODULATES WHOM
1 STACK (default)	4 › 3 › 2 › 1 (a single stack: the richest and most aggressive).
2 Y	(3 + 4) › 2 › 1
3 BRANCH	3 › 2 › 1 and 4 › 1
4 FORK	4 › 3 › 1 and 2 › 1
5 TWIN	2 › 1 and 4 › 3 (two pairs, two carriers).
6 FAN	4 › 1, 2 and 3
7 TRIPLE	4 › 3, with 1 and 2 on their own.
8 ORGAN	All 4 on their own (additive, like an organ).

CONTROL	DEFAULT	WHAT IT DOES
FEEDBACK	0%	OP 4 modulates itself: from sine to saw and all the way to noise.
FOLD	0%	Wavefolding at the output: harsh harmonics.
RING MOD	0%	Multiplies the carriers by OP 2's waveform: a metallic, broken-bell sound.
NOISE	0%	Adds noise (breath, attack).
DRIVE	50%	Up to halfway it's just level; above that it saturates before the filter.

STEP 4

4 The 4 operators



OPERATOR 1 to 4: each with its own color. The display shows the waveform, the ratio (×) and the role (OUT or MOD).

CONTROL	RANGE / DEFAULT	WHAT IT DOES
WAVE	SINE (default), HALF SINE, ABS SINE, PULSE SINE, ALT SINE, CAMEL, SQUARE, SAW	Operator waveform. Anything other than SINE starts out richer.
LEVEL	0–100% · OP 1 **100%**, others **0%**	On a carrier, it's the volume. On a modulator, it's the **amount of FM** (the brightness).
RATIO	×0.5, ×1 … ×16 · **×1**	Operator tuning relative to the played note.
FINE	+0 to +100% · **+0%**	Fine offset of the ratio: ratio = RATIO × (1 + FINE). This is where the metal comes from (×1.41, ×3.5, ×2.76…).
DETUNE	±50 ct · **0**	Detune in cents: adds beating and width.
VELOCITY	0–100% · **0%**	How much key velocity changes the operator level. On a modulator, playing harder makes the sound brighter.
ATTACK / DECAY / SUSTAIN / RELEASE	default 1 ms / 800 ms / 100% / 300 ms	Operator envelope (DECAY up to 20 s). On a modulator, it's the **timbre** envelope.

BELL RECIPE. Algorithm 1 STACK. OP 1 at LEVEL 100%. OP 2 with RATIO ×3 and FINE around +17% (×3.5), LEVEL halfway, short DECAY and SUSTAIN 0%. The attack turns metallic and the tail stays clean.

FM BASS RECIPE. Algorithm 5 TWIN or 1 STACK, modulator at ×1 with low LEVEL and a little FEEDBACK. Use the modulator envelope for the "pluck" of the attack.

STEP 5

5 Filters and envelopes



FILTERS

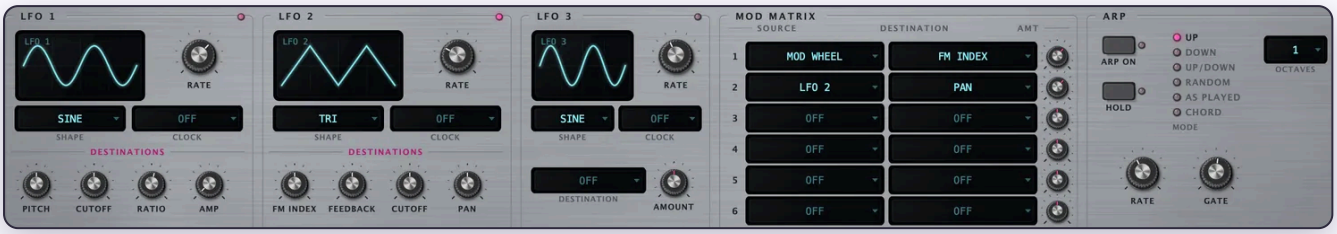
CONTROL	RANGE / DEFAULT	WHAT IT DOES
FILTER 1	**LP 24**, LP 12, HP, BP, BS, COMB, VOWEL	LP 24 is a ladder. COMB is a comb tuned to the cutoff (with KEY TRACK at 100% it plays the note: string, tube). VOWEL makes U-O-A-E-I vowels that move with the cutoff.
FILTER 2	**LP**, HP, BP, BS, COMB, VOWEL	Second filter.
CUTOFF	20 Hz–20 kHz · **20 kHz**	Filter 1 cutoff.
CUTOFF 2	±48 st · **0**	Filter 2 cutoff, in semitones relative to CUTOFF.
RESONANCE / ENV AMOUNT / KEY TRACK	RES 0% (F1), 20% (F2); ENV ±; KEY 0%	Boost at the cutoff, how much FILTER ENV opens (+) or closes (−) the filter, and how much the cutoff follows the note.
ROUTING / **BALANCE**	**SERIAL** / PARALLEL	Filters in series or in parallel; BALANCE blends the two.
SATURATION / **SAT DRIVE**	**OFF**, LIGHT, TUBE, HARD, FOLD, BITS, RATE · SAT DRIVE **30%**	Saturation between the two filters.

ENVELOPES

FILTER ENV (opens and closes the filters) and **AMP ENV** (volume): `ATTACK`, `DECAY`, `SUSTAIN`, `RELEASE`, `SLOPE` (the sustain slowly rises or falls while the key is held) and `VELOCITY` (how much key velocity counts). Each one has a display showing its shape.

STEP 6

6 Movement: LFOs, matrix and ARP



LFO 1, 2 and 3

Each LFO has `RATE`, `SHAPE` (SINE, TRI, SAW, SQUARE, S&H, S&G) and `CLOCK` (OFF = free, or 1/16 to 4 bars, locked to the project tempo). The destinations are pre-assigned:

LFO	DESTINATIONS
LFO 1	PITCH, CUTOFF, **RATIO** (the modulator ratios wobbling: the psychedelic effect) and AMP.
LFO 2	FM INDEX (the level of all modulators), FEEDBACK, CUTOFF and PAN.
LFO 3	One DESTINATION of your choice, with AMOUNT.

MOD MATRIX (6 slots)

Each slot connects a `SOURCE` (mod wheel, aftertouch, velocity, key, LFO 1/2/3, filt env, amp env, random) to a `DESTINATION` (cutoff, cutoff 2, resonance, pitch, FM INDEX, each operator's level, OP 2–4 ratio, feedback, fold, ring, noise, saturation, filter balance, pan, volume), with an `AMT` (±). Slot 1 comes preset to **MOD WHEEL › FM INDEX**: the mod wheel opens up the brightness of the sound.

ARP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ARP ON / **HOLD**	on/off	Turns the arpeggiator on; HOLD keeps the chord going after you release the keys.
MODE	**UP**, DOWN, UP/DOWN, RANDOM, AS PLAYED, CHORD	Note order.
OCTAVES	1–4	How many octaves the arpeggio covers.
RATE / **GATE**	1/32 to 1/4 (**1/16**) · GATE 5–100% (**55%**)	Speed and length of each note.

With the DAW playing, the arpeggio follows its tempo; when stopped, it uses its own clock at the project BPM.

STEP 7

7 VOICE and EFFECTS



VOICE	WHAT IT DOES
KEY MODE	**POLY** (default), MONO or LEGATO (mono with no retrigger between overlapping notes).
GLIDE	Pitch glide between notes (0 to 2 s).
BEND	Pitch bend range: 1 to 24 semitones.
UNISON / DETUNE / SPREAD	1 to 4 voices per note, detuned (DETUNE, default 25%) and spread across the stereo field (SPREAD, default 60%).
DRIFT	Each voice drifts out of tune a tiny bit, slowly, like vintage gear.
MASTER	Overall volume: −36 to +12 dB (default −6 dB).



EFFECTS	CONTROLS
DISTORTION	TYPE (SOFT, HARD, FUZZ, RECT) and AMOUNT.
ANALOG BOOST	AMOUNT and TUNE (40–400 Hz): low-end boost.
CHORUS / PHASER	MIX and DEPTH.
DELAY	SEND, TIME (synced to the project tempo) and FEEDBACK; ping-pong, getting darker.
REVERB	SEND, SIZE and DAMPING.
EQ	LOW (150 Hz), MID (1.2 kHz) and HIGH (6 kHz), ±12 dB.

Using the knobs and displays

- Hover over a knob and the label below it turns into the value. **Double-click** resets to the default.
- In the display fields (algorithm, wave, matrix, clock…): **click** to open the list; **drag vertically** or use the **mouse wheel** to change it.

STEP 8

8 Save your own presets

- `STORE` (or `Salvar como novo preset…`, save as new preset, in BROWSE) stores the sound in the **USER** list. Your presets can be overwritten, renamed and deleted (they go to the trash).
- `Importar banco de presets…` (import preset bank) installs a **.mrcbank** bank, which appears below FACTORY 1. `Remover banco` (remove bank) sends it to the trash.
- USER folders: Windows **%APPDATA%\InsideStudioSP\MERCURIO\Presets** · macOS **~/Library/Audio/Presets/InsideStudioSP/MERCURIO**.

FAQ and troubleshooting

SITUATION	WHAT TO DO
No sound	Check the license (BROWSE › Licença…): until activated, the instrument stays silent. Check that the track receives MIDI (MIDI LED) and the MASTER level.
I tweak an operator and nothing changes	Check in the display whether it's OUT or MOD and whether its LEVEL is above 0%. In algorithm 8 ORGAN nobody modulates anybody.
Sound too harsh or noisy	Lower the modulators' LEVEL, FEEDBACK, FOLD or DRIVE.
Stuck notes playing on their own	The ARP has HOLD on. Turn HOLD off.
High CPU	Reduce UNISON and the number of simultaneous notes.
Window too small	BROWSE › window size (60–180%). It's saved with the project.

30-second summary

1. MERCÚRIO on an instrument track → activate the license.
2. **BROWSE** → pick a category and step through the presets.
3. Change the **ALGORITHM** and hear the difference.
4. On the modulators, tweak **LEVEL**, **RATIO** and **FINE**.
5. Use the **mod wheel** (already routed to FM INDEX) and the **LFOs** to add movement.
6. Finish with filter, effects and **STORE**.

> **SUPPORT.** Questions, suggestions or issues: **contato@insidestudiosp.com.br** · **insidestudiosp.com.br**